Transcriptional Noise and Rare Phenotypic Switching in Epithelial–Mesenchymal Heterogeneity

Presented by:
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Dr. Paras Jain (Post Doc)
Cancer Systems Biology lab,
Dep of Bioengineering, Indian Institute of Science Bengaluru, India

Organised by

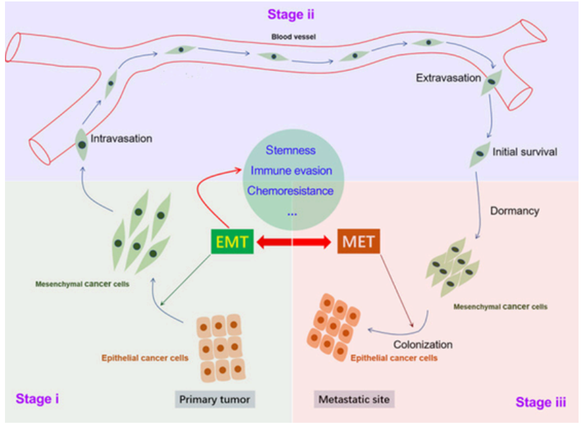
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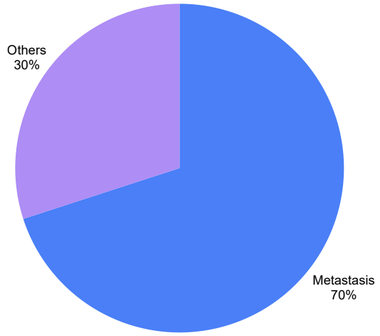
Research Catalyst Global Symposium 2026

BACKGROUND

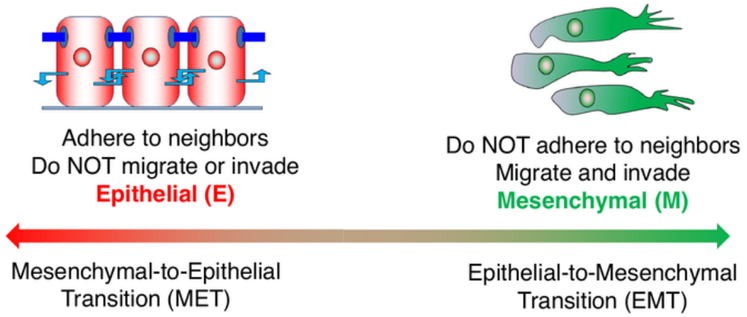
Why metastasis needs to be addressed?



Cells can separate from the primary organ, enter the circulatory and lymphatic system, and invade other organs



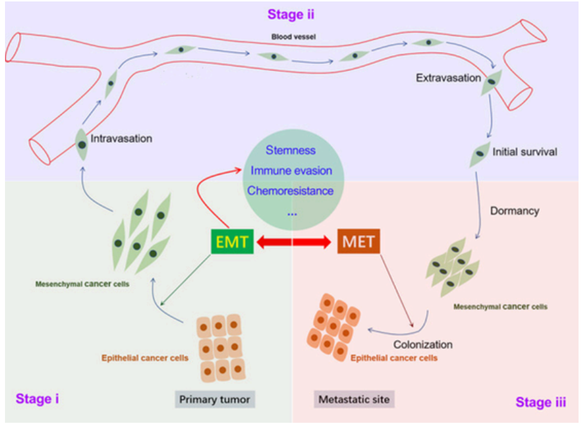
Metastasis accounts for 70-90% of solid tumor related deaths



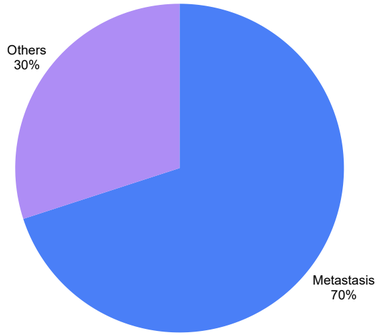
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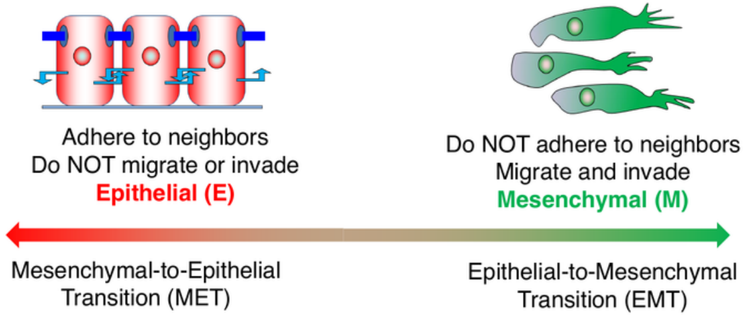
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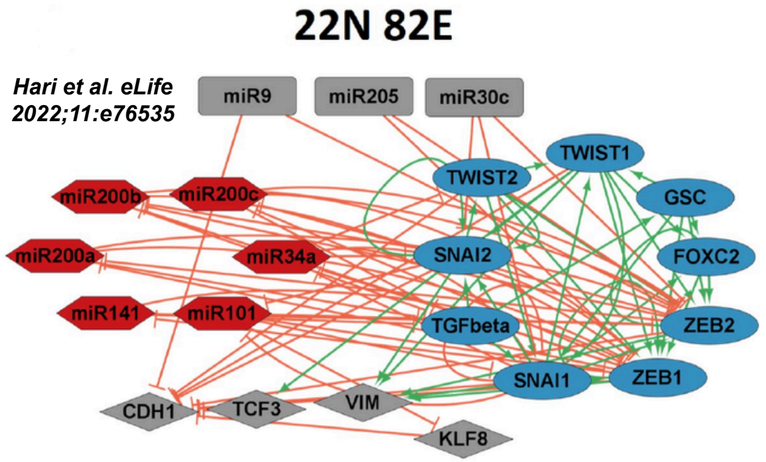


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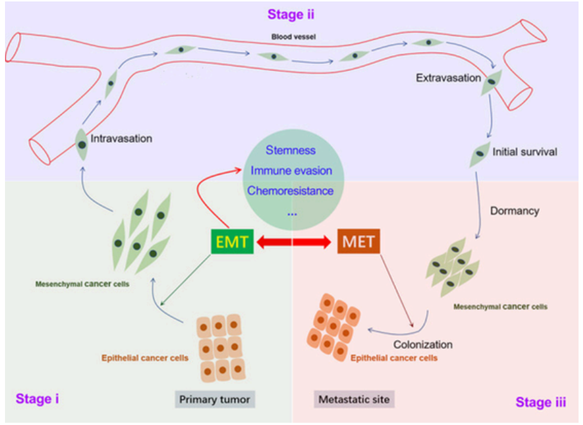


EMP network with 22 Nodes & 82 Edges

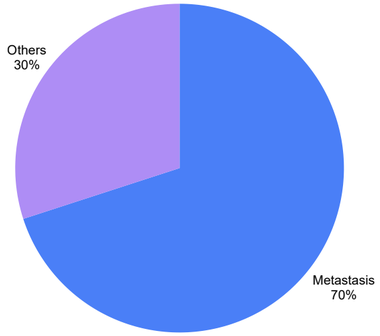
- TEAMS network regulates epithelial mesenchymal plasticity
- Boolean simulations suggest that network is robust to external perturbations
- Terminal states are stable and hybrid states are vulnerable to perturbations

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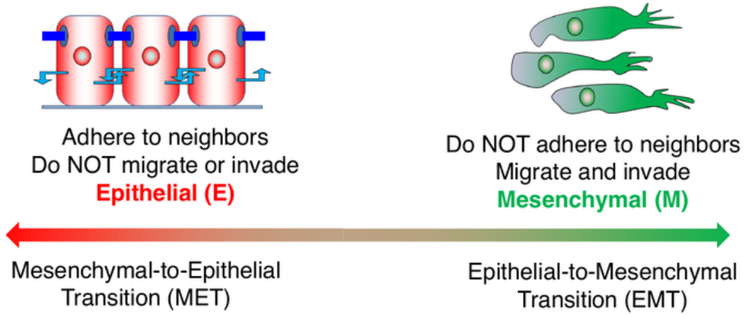
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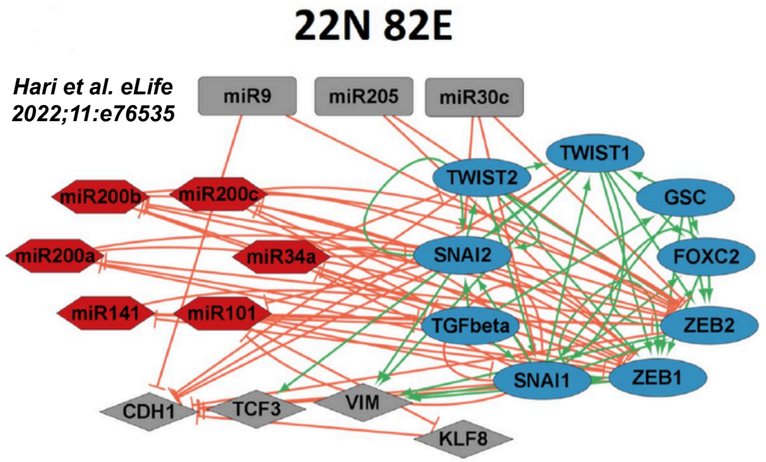


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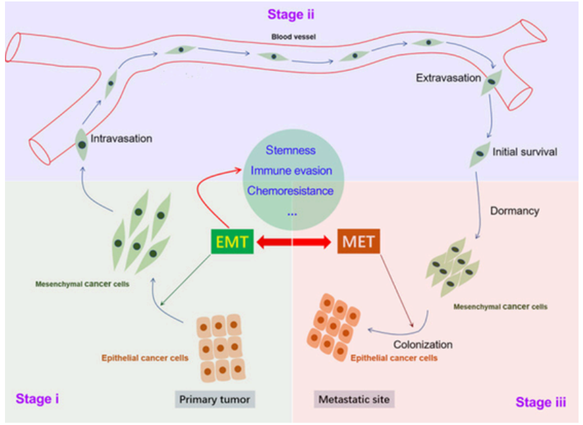
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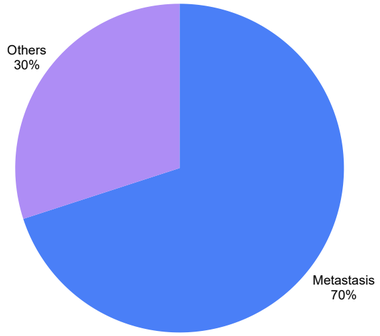
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- There must be SOMETHING that breaks the robustness of the network and makes it vulnerable to perturbations
- How can cells spontaneously switch between terminal & hybrid phenotypic states despite **robustness** of the underlying network?

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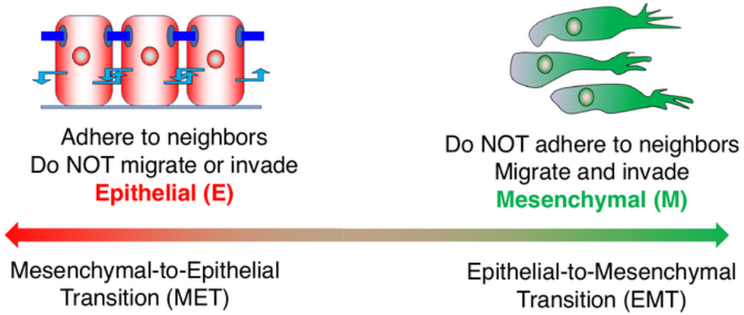
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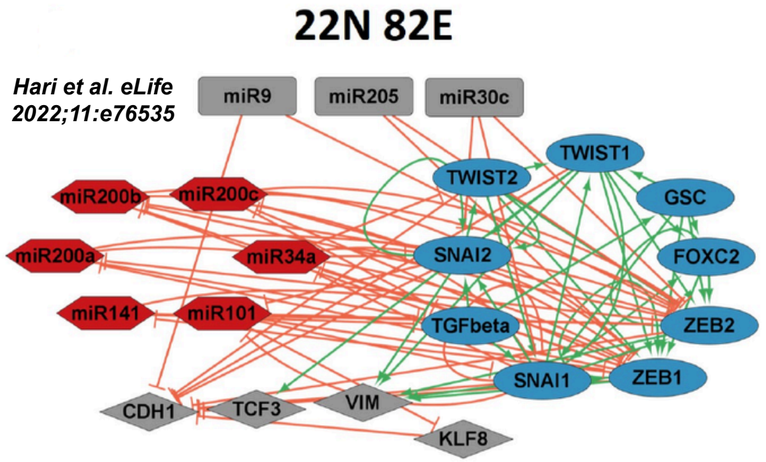


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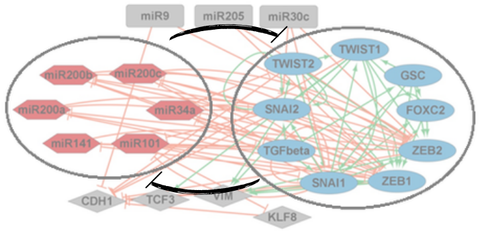
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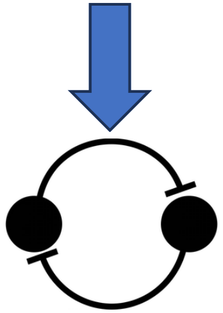
HYPOTHESIS

- **Network size and connectivity influence the emergence of heterogeneity.**
- **Transcriptional noise may contribute to spontaneous switching**

Computational Framework & Methodology



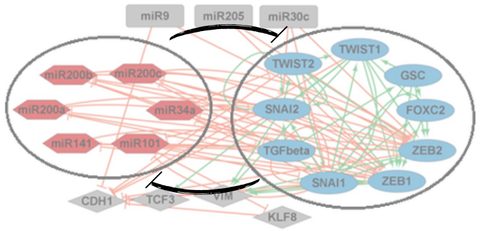
Simplified



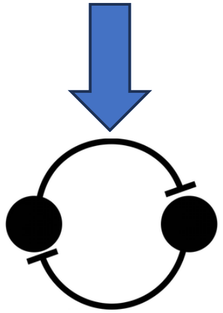
Toggle switch

- Toggle switch is a simplified representation of the “teams” network
- Regulation → Shifted hill function
- Gaussian noise assumption
- Std. dev. is a function of copy number

Computational Framework & Methodology



Simplified



Toggle switch

production degradation

$$\frac{dm_1(t)}{dt} = \left\{g \cdot \left(1-(1-\lambda)\cdot H(m_2)\right) + \sigma_{birth,1}\cdot N(0,1)\right\} - \left\{k\cdot m_1 + \sigma_{death,1}\cdot N(0,1)\right\}$$

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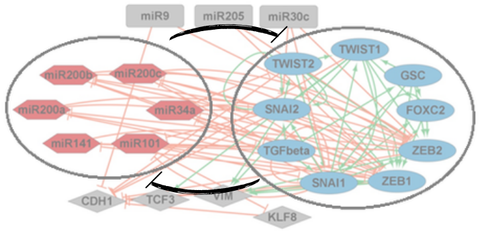
$$\sigma_{birth,i} = \sqrt{g\cdot\left(1-(1-\lambda)\cdot H\left(m_j\right)\right)} \qquad H(m) = \frac{m^n}{m^n + K^n}$$

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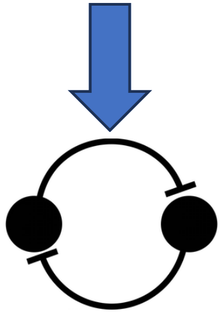
Chemical Langevin Equation

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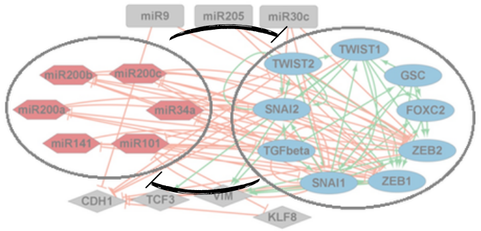
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Toggle switch → **Chemical Langevin Equation** → **Parameters generated using RACIPE** →

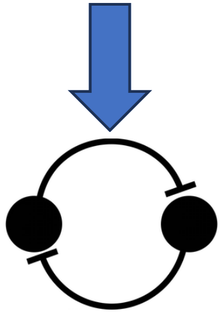
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- Threshold (λ) → using half-functional rule
- Only bistable parameters considered
- ODE simulation to obtain steady states

Computational Framework & Methodology



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Toggle switch → **Chemical Langevin Equation** → **Parameters generated using RACIPE** → **Simulation & Statistical Analysis**

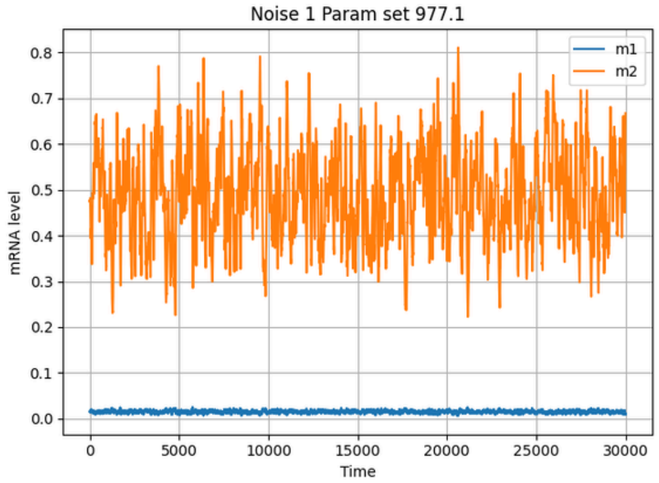
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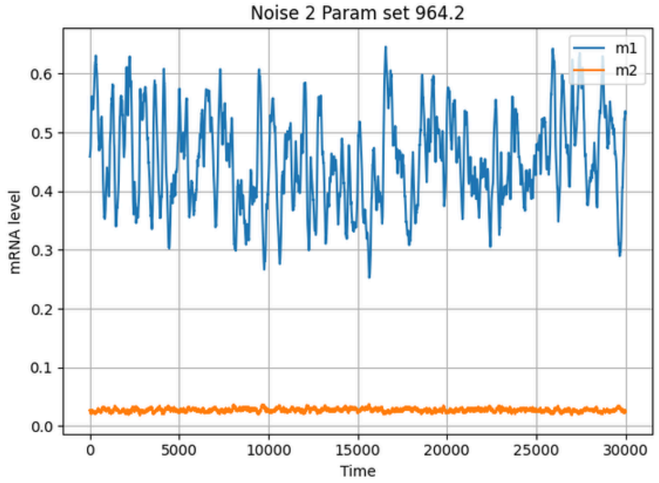
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- Noise in one nodes vs both nodes
- 30,000 time points
- Calculated Jensen Shanon Distance
- Distribution of relative expression

Results I: Expression Dynamics Under Transcriptional Noise

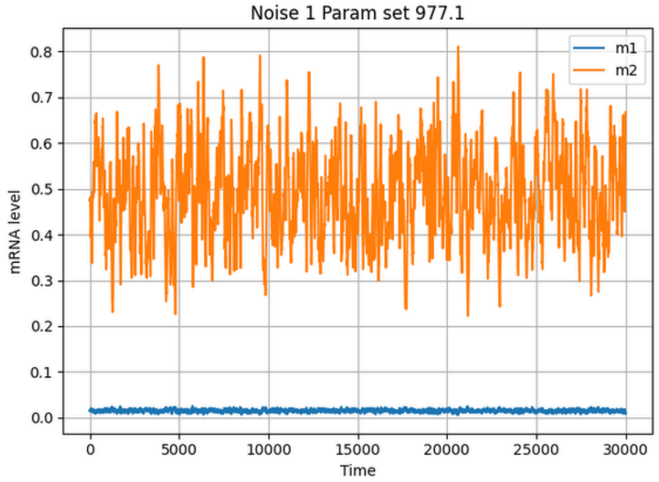
I. Terminal states maintained

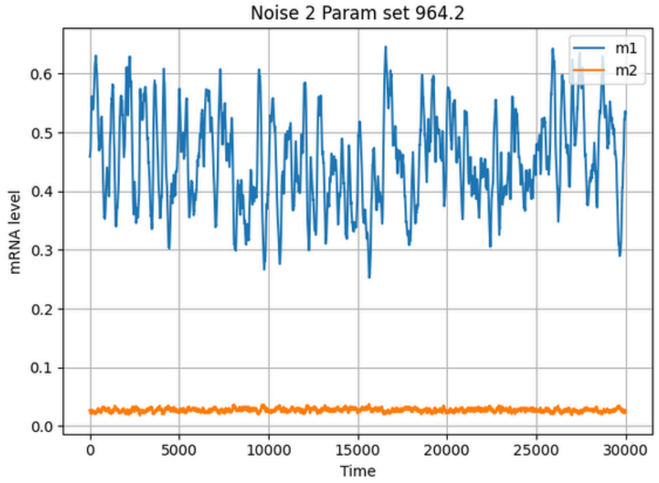




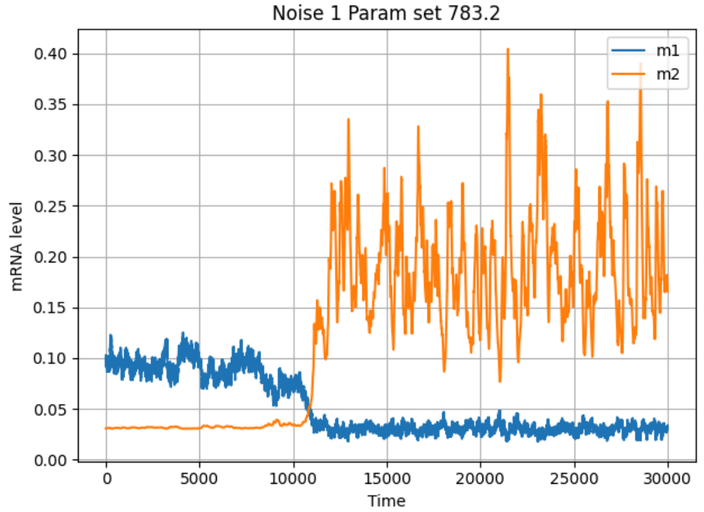
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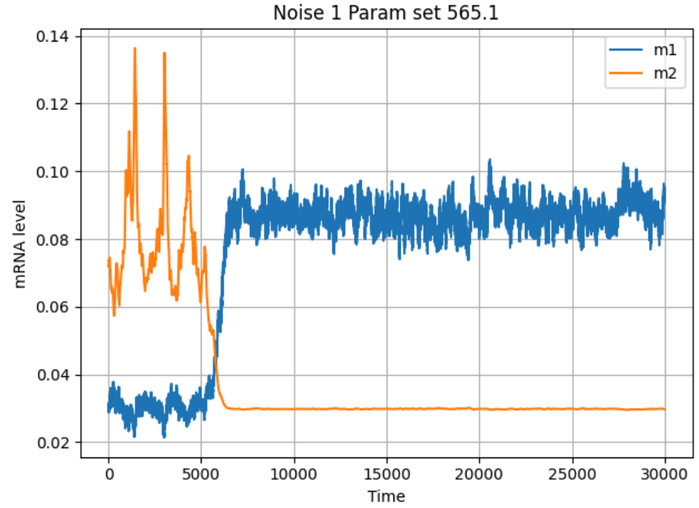
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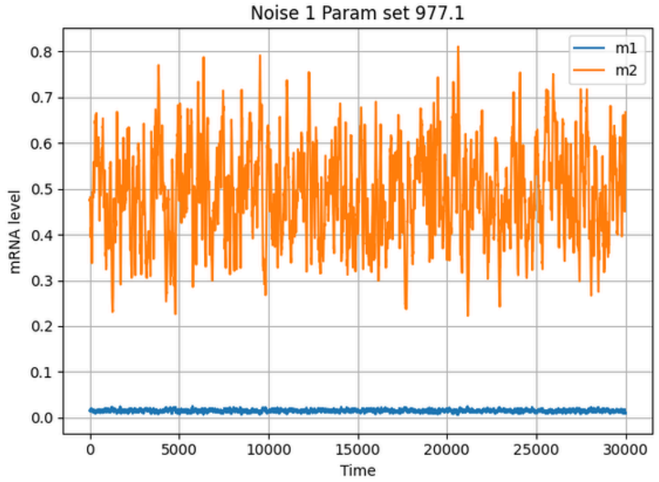
II. Single switching event

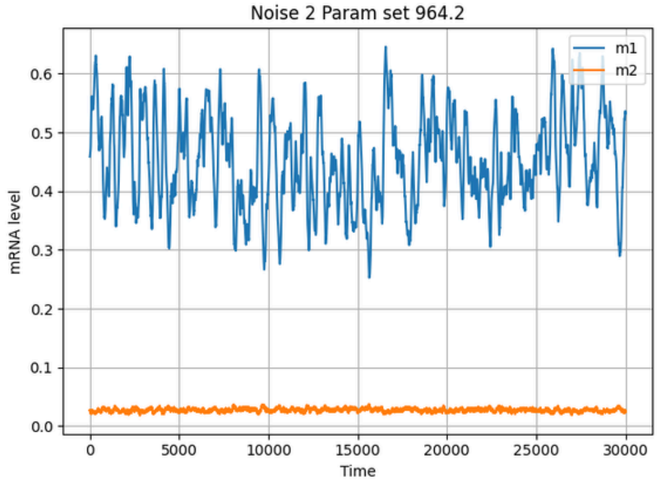




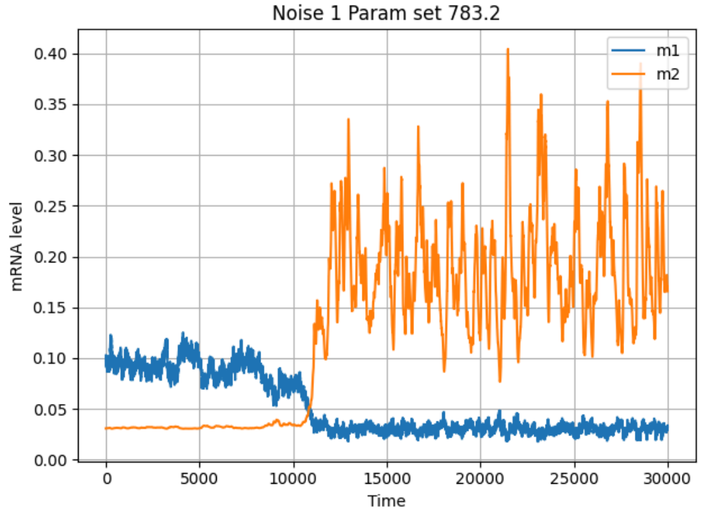
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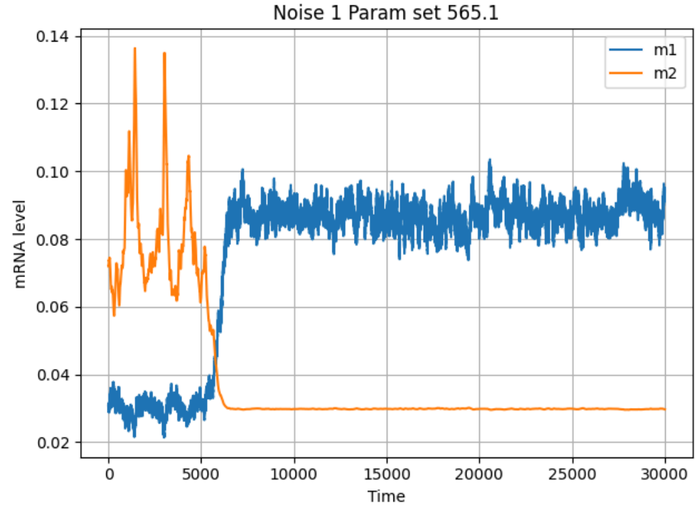
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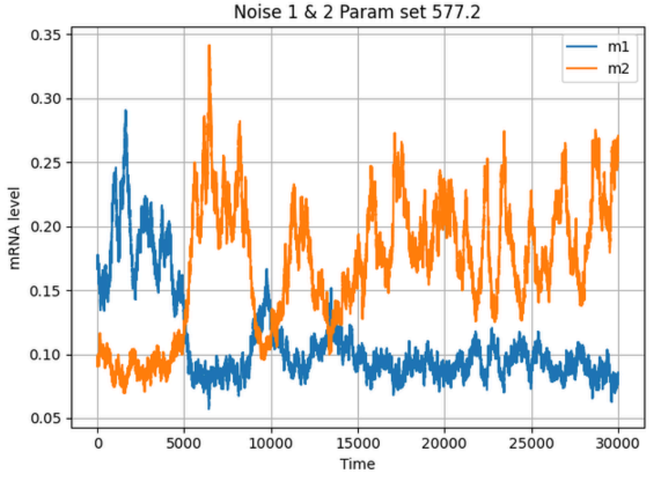


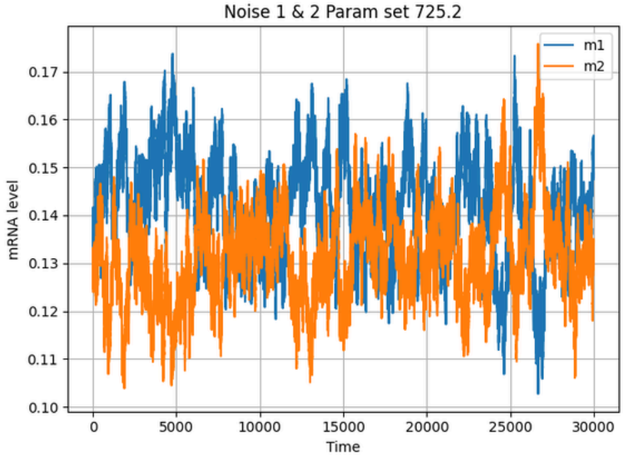
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III. (RARE) spontaneous switching (Hybrid states)

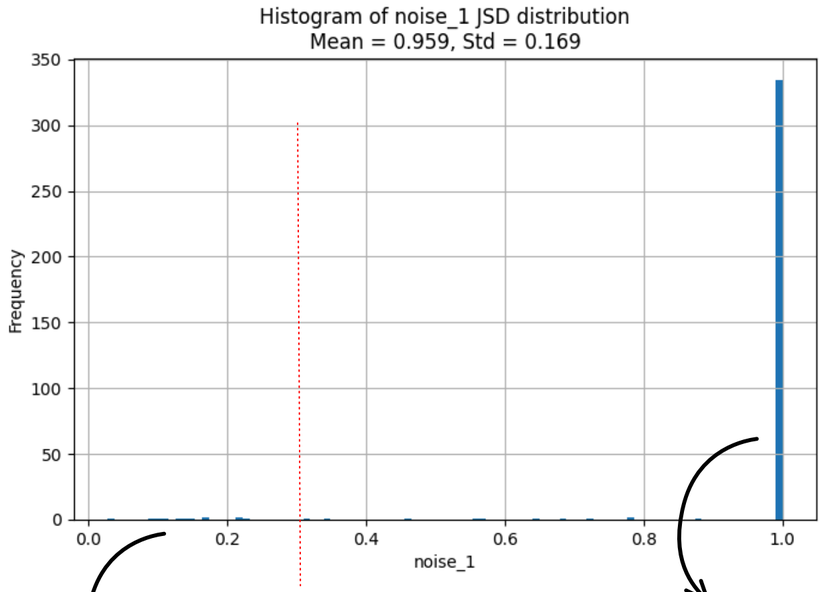




Results II: Statistical Evidence of Hybrid States

Are the expression dynamics dependent on the initial conditions??

High JSD – Distinct distributions (dependent)
Low JSD – Overlap b/w distributions (independent)



majority of parameter sets show high JSD

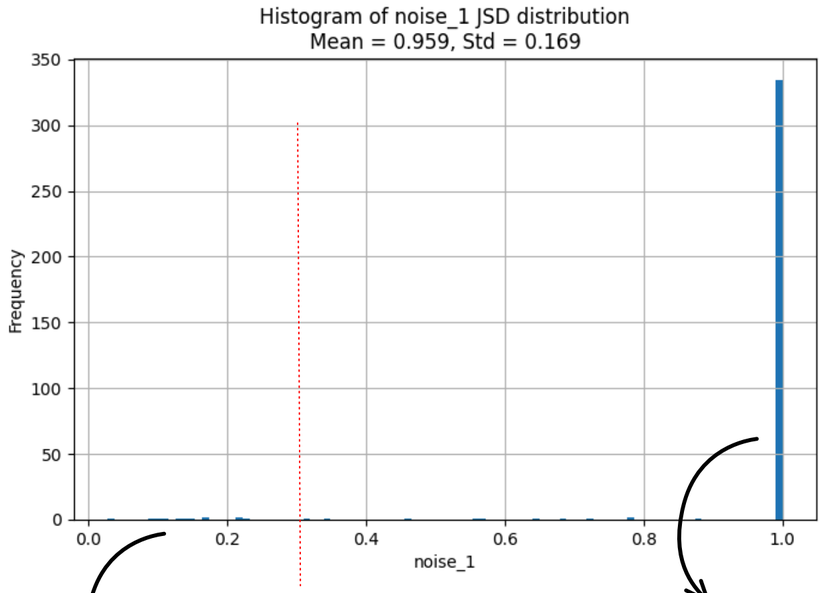
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This is not necessarily an evidence of hybrid states, but that hybrid states could be possible under certain regimes

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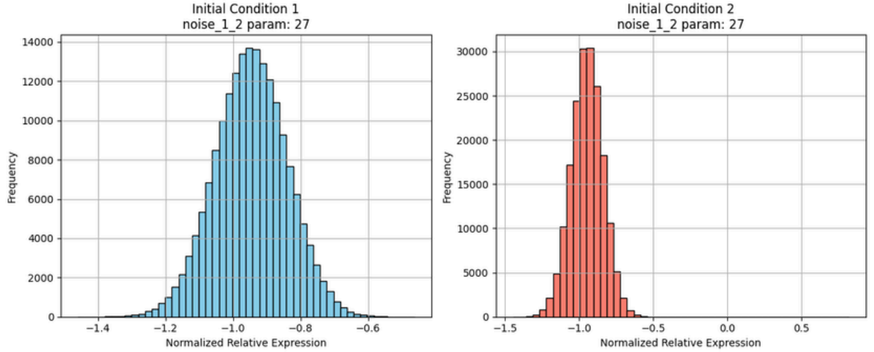
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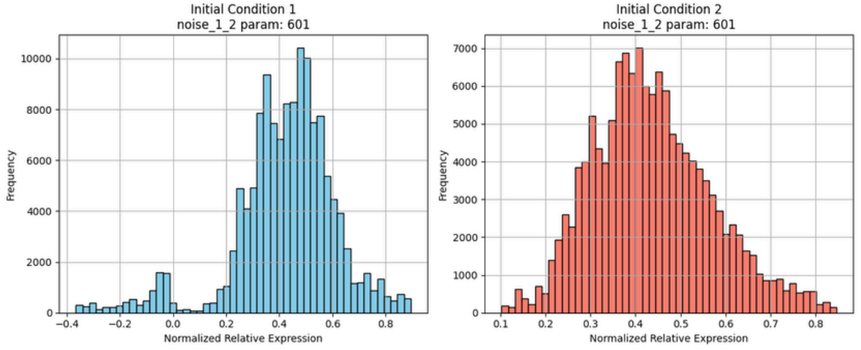
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Relative expression analysis (norm. node1 exp – norm. node2 exp)

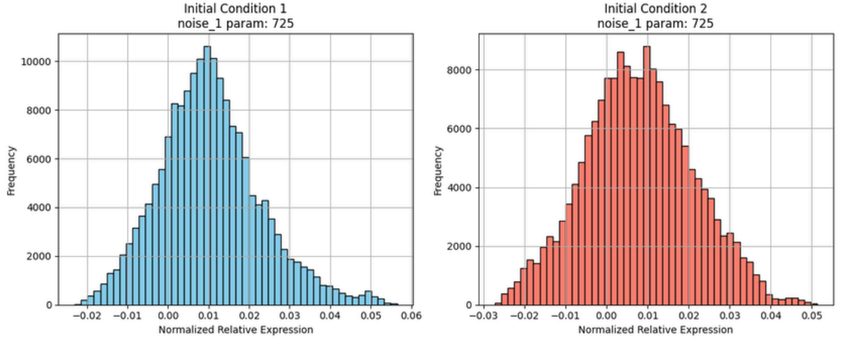
Unimodal distribution
(terminal states maintained)



Bimodal distribution
(switching of terminal states)



Hybrid states
(equal expression
by both nodes)



Conclusions & Future Directions

Key Findings

- Stochastic simulations reveal multiple expression patterns under noise
- Noise-associated switching can occur in simplified toggle-switch.
- Hybrid state expression patterns are very rare and highly dependent on the model parameters

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Future Directions

- More **rigorous methods** for statistical analysis
- Extend analysis to **larger** and more **complicated networks**
- Model different noise mechanisms
- Validate predicted switching behaviour against experimental single cell gene-expression data.

Thank You!

Questions ??

Feedback ??